

KUNAL HEMANT PASAD

Artificial Intelligence Engineer | Generative AI | LLM Systems | Multimodal AI

60 Paterson St, New Brunswick, NJ 08901

📞 848-391-7620 ✉ kunalpasad6523@gmail.com 🌐 [Portfolio](#) [Linkedin](#) [Github](#)

EDUCATION

Rutgers, the State University of New Jersey, New Brunswick, United States

Expected: May 2027

Master of Science in Computer Science

GPA: 3.83/4.0

Relevant Coursework: Machine Learning, Distributed Systems, Natural Language Processing, Database Systems

Sardar Patel Institute of Technology, Mumbai, India

Dec 2021 – May 2025

Bachelor of Technology in Computer Engineering

GPA: 8.62/10

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Artificial Intelligence

TECHNICAL SKILLS

Languages: Python, SQL

AI & LLMs: Generative AI, LLMs, RAG, Prompt Engineering, Fine-tuning (QLoRA), Multimodal AI, LLM Evaluation

Frameworks: PyTorch, HuggingFace Transformers, FastAPI

Data Systems: Vector Databases, Embeddings (FAISS), ETL Pipelines, Feature Engineering

Cloud & Tools: AWS (S3, EC2, Lambda), Docker, Kubernetes, CI/CD

Concepts: Model Evaluation, Retrieval Systems, NLP, Distributed Systems, Inference Optimization

PROFESSIONAL EXPERIENCE

EternIQ

Jan 2025 – May 2025

Machine Learning Engineer Intern

Mumbai, India

- Engineered scalable LLM inference pipelines for structured reasoning and code generation in production
- Designed prompt engineering strategies reducing hallucination rates by approximately 25% across responses
- Developed evaluation frameworks analyzing reasoning accuracy across 1K+ outputs improving reliability and consistency
- Built data pipelines processing 100K+ records with validation, normalization, and schema enforcement

Mukesoftware IT Consultants

Jul 2023 – Feb 2024

Full Stack Web Developer Intern

Mumbai, India

- Developed backend APIs using Laravel supporting scalable applications and efficient data-driven workflows in production
- Optimized SQL queries and indexing strategies reducing API latency by 30% in high-load environments
- Processed and managed 50K+ relational records improving database performance and ensuring efficient data retrieval
- Implemented validation and integrity checks ensuring consistent and reliable backend data processing systems

PROJECTS

InstantStory – Multimodal LLM Story Generation System | Python, LLMs, QLoRA, FastAPI

Mar 2026

- Engineered end-to-end LLM pipeline using Mistral-7B improving narrative coherence by approximately 22% over baseline models
- Fine-tuned language model using QLoRA reducing training memory usage by nearly 60% without performance loss
- Deployed FastAPI inference pipeline reducing latency by 25% through batching and optimized prompt strategies

MindSync – AI Knowledge Assistant | RAG, FastAPI, AWS, Vector Search

Oct 2025

- Architected retrieval-augmented generation pipeline enabling grounded responses over unstructured documents reducing hallucinations
- Implemented semantic vector search and context-aware prompting reducing incorrect responses by approximately 25% in evaluations
- Optimized retrieval and inference pipeline reducing latency by nearly 30% using caching and chunk selection

Multilingual Sports QA Benchmark (EMNLP 2025) | LLMs, Multilingual NLP

Nov 2025

- Co-developed multilingual QA benchmark with 33K+ questions spanning 60 countries and 11 languages for LLM evaluation
- Designed multi-hop reasoning templates generating 30K+ samples to evaluate compositional reasoning capabilities across models
- Conducted large-scale evaluation revealing up to 18% performance drop in cross-lingual reasoning tasks

Deepfake Guard – Audio Deepfake Detection System | PyTorch, CNNs, MFCC

May 2025

- Developed CNN-based deepfake detection system achieving approximately 93% accuracy using MFCC features
- Engineered feature extraction pipelines converting raw audio into spectrograms improving robustness and generalization capabilities
- Integrated explainability techniques including SHAP and Grad-CAM to interpret outputs and highlight features